

ARUN ARUNACHALAM

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PROFESSIONAL SUMMARY

Data platform leader with 17+ years architecting and delivering enterprise-scale data engineering and analytics platforms in highly regulated industries, with deep expertise in cloud-native architecture, event-driven systems, and AI-enabled data pipelines. Proven record modernizing legacy data infrastructure into scalable, high-availability platforms that drive measurable business outcomes across complex, matrixed organizations. Combines hands-on engineering depth in Spark, Databricks, Kafka, and AWS with applied GenAI (RAG pipelines, agentic AI) to accelerate intelligent data product delivery. Experienced navigating regulatory and compliance requirements while partnering with executive stakeholders to align data platform strategy with enterprise priorities.

SIGNATURE IMPACT

- **\$8M** enterprise transformation → **40% performance improvement, 30% cost reduction**
- **AI-assisted engineering** → **35% faster delivery, 45% productivity gain**
- Scaled platforms from 10 to **50+ systems** with **99.97% uptime**
- Enabled **~\$2B capital efficiency** via risk platform transformation

CORE CAPABILITIES

Data Engineering & Platform Architecture (Spark, Databricks, ETL/ELT) | Cloud-Native Architecture (AWS, Kubernetes) | Event-Driven Systems (Kafka) | AI & GenAI Platforms (RAG, Agentic AI) | ML & Anomaly Detection | Snowflake, Hadoop, Cassandra, MongoDB | Trading & Risk Data Platforms | Platform Modernization & Legacy Migration | Regulatory & Compliance Systems (SOX, FRB, FINRA, SEC) | Executive Stakeholder Leadership | Agile & SDLC Delivery

PROFESSIONAL EXPERIENCE

Bank of America — Director, Global Risk & Capital Markets Technology (2015 - Present)

- Led enterprise-scale data platform transformation supporting **\$1T+ daily processing**, modernizing legacy data pipelines into cloud-native, event-driven architecture (AWS, Kafka, Kubernetes)
- Delivered real-time data ingestion, pricing, and risk analytics platforms improving throughput by **40%** and reducing infrastructure costs by **30%**
- Architected and implemented ETL/ELT data engineering workflows using Spark and Databricks, enabling high-volume, low-latency data delivery across trading and risk systems
- Implemented AI-assisted development workflows and ML-based anomaly detection to improve data quality and operational intelligence
- Scaled data platforms from 10 to **50+ integrated systems** while maintaining **99.97% uptime** across mission-critical pipelines
- Ensured regulatory compliance (FRB, FINRA, SEC, SOX) across all data platform components and reporting systems
- Partnered with executive stakeholders across technology and business to align data platform roadmap with enterprise strategy

- Drove **15% engineering velocity improvement** through Agile delivery practices and SDLC optimization

Bank of America — VP, Structured Products & Mortgage Technology (2011 - 2015)

- Built real-time pricing and analytics data platforms for structured products and mortgage trading, contributing to **~\$2B capital efficiency**
- Led consolidation of disparate risk data systems into unified, high-availability platforms, reducing operational complexity and improving data consistency
- Delivered end-to-end data pipelines supporting analytics and reporting for structured product portfolios

Bank of America — AVP, Whole Loan Trading Desk Support (prior to 2011)

- Architected and delivered end-to-end Java EE data and trading support applications for the whole loan trading desk
- Introduced CI/CD practices reducing release cycle from monthly to weekly, improving data platform delivery cadence

Senior Programmer Analyst — Knight Trading (2004)

DB2 / WebSphere Administrator & Sr. Java Developer — Kyocera Mita (2003)

Sr. Java Web Developer — Mellon Investor Services (2001)

Sr. Java GUI Developer — Thomson Financials (2001)

Sr. Java GUI Developer — Icor Brokerage (2000)

AI & INNOVATION

- Building GenAI platforms (RAG pipelines, agentic AI systems) for real-world data and operational applications
- Applied ML for anomaly detection and operational intelligence across large-scale data pipelines
- Driving AI-assisted SDLC workflows improving data platform delivery speed and quality

EDUCATION

Bachelor of Technology — Anna University

SKILLS

Architecture: Cloud-native architecture (AWS, Kubernetes), enterprise data platform architecture, event-driven systems (Kafka), distributed systems, high-availability platforms (99.97% uptime), ETL/ELT pipeline design **AI & Data:** GenAI (RAG pipelines, agentic AI), ML anomaly detection, AI-assisted SDLC, Spark, Databricks, Snowflake, Hadoop, Cassandra, MongoDB **Engineering:** Java, Python, Node.js, SQL, Oracle, Sybase, REST/SOA, WebLogic, Kafka, Kubernetes, AWS **Leadership:** Executive stakeholder management, strategic planning, program management, SDLC, Agile, ITIL, SOX, FRB/FINRA/SEC compliance, P&L ownership **Tools:** ServiceNow, JIRA, Git, Visio, Microsoft Project